



Distinct Pattern of Solar Modulation of Galactic Cosmic Rays above a High Geomagnetic Cutoff Rigidity

Pierre-Simon Mangeard¹, John Clem¹, Paul Evenson¹, Roger Pyle²,
Warit Mitthumsiri³, David Ruffolo³, Alejandro Sáiz³, and Tanin Nutaro⁴

¹ Bartol Research Institute and Department of Physics and Astronomy, University of Delaware, Newark, DE 19716, USA

² Pyle Consulting Group, Inc., St. Charles, IL 60174, USA

³ Department of Physics, Faculty of Science, Mahidol University, Bangkok 10400, Thailand; david.ruf@mahidol.ac.th

⁴ Department of Physics, Faculty of Science, Ubon Ratchathani University, Ubon Ratchathani 34190, Thailand

Received 2018 March 23; revised 2018 April 5; accepted 2018 April 8; published 2018 May 3

Abstract

Solar modulation refers to Galactic cosmic-ray variations with the ~ 11 yr sunspot cycle and ~ 22 yr solar magnetic cycle and is relevant to the space radiation environment and effects on Earth's atmosphere. Its complicated dependence on solar and heliospheric conditions is only roughly understood and has been empirically modeled in terms of a single modulation parameter. Most analyses of solar modulation use neutron monitor (NM) data from locations with relatively low geomagnetic cutoff rigidity, i.e., the threshold for cosmic rays to penetrate Earth's magnetic field. The Princess Sirindhorn Neutron Monitor at Doi Inthanon, Thailand, has the world's highest cutoff rigidity (≈ 17 GV) where observations span a complete solar modulation cycle (since late 2007). The pattern of solar modulation at Doi Inthanon during 2011–2014 was qualitatively very different from that at a low geomagnetic cutoff and is not well described by the same modulation parameter. At other times, NM count rates from Doi Inthanon and McMurdo, Antarctica (cutoff ~ 1 GV), were linearly correlated and confirm the observation from latitude surveys in the previous solar cycle that the slope of the correlation changes with solar magnetic polarity. Low solar magnetic tilt angles ($< 40^\circ$ at negative polarity) were well correlated with variations at both NM stations, as predicted by drift models. At a higher tilt angle, the Doi Inthanon count rate is well correlated with the interplanetary magnetic field, which is consistent with an increase in diffusion at high rigidity short-circuiting the effects of drifts and the heliospheric current sheet.

Key words: cosmic rays – solar–terrestrial relations – Sun: activity – Sun: magnetic fields

1. Introduction

While Galactic cosmic rays (GCRs) originate outside the heliosphere, the time variation of the flux traversing the solar wind to arrive at or near Earth is produced by solar and heliospheric effects. As a result, the GCR flux below ~ 50 GeV varies throughout the sunspot (solar activity) cycle of ~ 11 yr (Forbush 1954) and the solar magnetic cycle of ~ 22 yr (Jokipii & Thomas 1981), a process known as solar modulation. The GCR flux is highest at sunspot minimum and lowest at sunspot maximum. This process affects the space radiation environment and ionization in Earth's atmosphere (Bazilevskaya et al. 2008). Observations of solar modulation also provide important and arguably unique information about the global properties of the heliosphere, because GCRs reach Earth after traveling over a great range of distance, heliolatitude, and heliolongitude. Their transport is affected by the interplanetary magnetic field (IMF), magnetic turbulence, and solar wind speed over a large heliospheric volume.

Direct measurements of cosmic rays have the advantage of determining the particle species and energy and hold great promise for the future. To date, energy-dependent spectra or abundance ratios at GeV-range energies have been reported for discrete epochs by using balloon-borne instruments, e.g., Clem & Evenson (2009) for electrons and positrons and Abe et al. (2016) for protons and alphas; by the *PAMELA* spacecraft for protons over four epochs during 2006–2009 (Adriani et al. 2013) and alphas over 2006–2008 (Adriani et al. 2011); and by the AMS-02 instrument on the *International Space*

Station over 2011–2013 for protons (Aguilar et al. 2015a) and alphas (Aguilar et al. 2015b).

Spectral variations of GCRs have also been tracked over discrete epochs by repeated latitude surveys using a neutron monitor (NM) on a ship. An NM can have a collecting area of tens of m^2 and can achieve much greater statistical accuracy over a much shorter time duration by detecting atmospheric showers from cosmic-ray ions. While the primary cosmic-ray species and energy are no longer directly measured, ground-based detectors have a minimum threshold in rigidity, $P = pc/q$, where p is the particle momentum, c is the speed of light, and q is the particle charge. The rigidity threshold is either a geomagnetic cutoff rigidity, P_c , which varies from near zero in polar regions to ~ 17 GV in parts of Southeast and South Asia, or an atmospheric cutoff (which for an NM is at ~ 1 GV) corresponding to the energy needed to generate secondary shower particles that can be detected at ground level (Mangeard et al. 2016a, 2016b). Therefore, the negative rate of change of the NM count rate with cutoff rigidity, as the ship travels, is proportional to a cosmic-ray spectrum at that rigidity, and minute changes in the spectrum can be determined by comparing results from successive latitude surveys. From a recent sequence of 13 consecutive latitude surveys during 1994–2007, Nuntiyakul et al. (2014) confirmed a crossover between spectra before and after the 2000 reversal in solar magnetic polarity. Furthermore, they reported that in years other than 2000, there were linear trends between the shipborne NM count rate at each cutoff rigidity and the fixed NM count rate at McMurdo, Antarctica, at low cutoff, with a change of slope upon the magnetic polarity reversal.

Continuous, high-precision measurements of the solar modulation of GCRs available from fixed NM stations are the subject of the present work. Each NM count rate is an integral measurement above a rather sharp cutoff rigidity, as described above, and the sensitivity (count rate per primary cosmic-ray flux) as a function of rigidity, known as the “yield function,” can be estimated by Monte Carlo simulation of particle interactions in the atmosphere and the detector (e.g., Mishev et al. 2013; Mangeard et al. 2016a, 2016b) or inferred from latitude survey and direct spectral measurements (Caballero-Lopez & Moraal 2012; Mitthumsiri et al. 2017). At low cutoff rigidity, there have been continuous measurements of solar modulation for six solar cycles (Simpson 2000; Oh et al. 2013).

These detailed time series have defied simple explanation. One popular model is the spherically symmetric force-field model, in which cosmic-ray spectra at Earth are characterized by a single modulation parameter, ϕ (Gleeson & Axford 1968; Caballero-Lopez & Moraal 2004). This parameter approximates the change in rigidity due to adiabatic deceleration after a GCR diffuses inward from an outer modulation boundary to the location of interest, assuming a near balance between the inward diffusive flux and the outward convective flux due to the solar wind. Based on NM observations at low cutoff rigidity, empirical modulation parameters have been tabulated as a function of time (Ghelfi et al. 2017b; Usoskin et al. 2017). Because of its simplicity, the force-field model remains popular for estimating cosmic-ray spectra as a function of solar modulation. Recent works (Corti et al. 2016; Gieseler et al. 2017) proposed modification of the force-field model by adding a second modulation parameter that leads to a rigidity-dependent solar modulation. While remaining relatively simple with only two parameters, these models give a better estimate of the solar modulation at low rigidities that are only marginally accessible with the NM network. In addition, there are observed effects of the solar magnetic polarity, commonly expressed as “ $A > 0$ ” when the solar magnetic field is predominantly outward in the northern hemisphere and inward in the southern hemisphere and “ $A < 0$ ” for the reverse polarity. Guiding center drifts of the charged cosmic rays in the IMF (Levy 1976; Jokipii et al. 1977) were proposed as a mechanism to explain changes in the time profiles of NM count rates upon a solar magnetic polarity reversal (Jokipii & Thomas 1981). Another proposed mechanism involves the effect of magnetic helicity on GCR diffusion (Bieber & Burger 1990). Both effects should depend on the product qA , and charge-sign-dependent modulation has been observed (Garcia-Munoz et al. 1986; Clem et al. 2000; Asaoka et al. 2002; Clem & Evenson 2009; Adriani et al. 2016), but charge sign and energy dependences are not yet fully resolved.

So far, most reported analyses of solar modulation based on NM time series were for a relatively low cutoff rigidity; there are not many long-term time series of NM data at $P_c > 12$ GV, as shown in Figure 1. Here we express P_c using the apparent cutoff rigidity, which takes obliquely incident GCRs into account (Clem et al. 1997) and has been calculated by weighting effective cutoffs calculated for nine arrival directions (Bieber et al. 1997). The effective cutoff is determined using the International Geomagnetic Reference Field (IGRF-12) and the methods of Lin et al. (1995). The apparent cutoff, P_c , changes as a function of time due to secular changes in Earth’s magnetic field. For NM stations worldwide with $P_c > 9.5$ GV,

Figure 1 shows the time periods and P_c values for which data are available, e.g., at the World Data Center for Cosmic Rays⁵ or the Neutron Monitor Database⁶ (NMDb). Vertical lines indicate approximate times of solar magnetic polarity reversal, which occurs near sunspot maxima. The NMs are sensitive to relativistic GCR ions with a positive charge, so Figure 1 also indicates time periods when positive magnetic polarity $A > 0$ resulted in qA -positive solar modulation and $A < 0$ resulted in qA -negative solar modulation. There are few NM stations at high P_c with data that span a solar cycle.

Among these, only the Princess Sirindhorn Neutron Monitor (PSNM) at Doi Inthanon, Thailand, had $P_c > 17$ GV, and indeed, it has the highest cutoff of any monitor currently in operation. After the start of full operation in 2007 December, the count rate rose to a maximum in 2009 (near sunspot minimum), fell to a minimum in 2014–2015 (near sunspot maximum), and then returned to its original count rate, thus spanning a full cycle. This time period also included a solar magnetic polarity reversal from $A < 0$ to $A > 0$.

In this work, we show that the solar modulation of GCRs as observed at Doi Inthanon from 2007 December to 2017 January exhibited a qualitatively distinct pattern from that observed at low cutoff rigidity. We also demonstrate that a force-field model using an empirical modulation parameter $\phi(t)$ as determined from low cutoff NM data provides a poor description of the time profile measured at Doi Inthanon. In other words, the spectral effects of solar modulation over roughly 1–17 GV are too complex to be well described by a single modulation parameter. We compare the Doi Inthanon NM count rate profile with low- P_c data from McMurdo, as well as the sunspot number, tilt angle of the heliospheric current sheet (HCS), solar wind speed, and IMF, and we identify different patterns of solar modulation during periods of low and high tilt angle.

2. PSNM Data and Water Vapor Correction

We focus on data from the PSNM at the summit of Doi Inthanon, Thailand’s highest mountain, with geographic coordinates 18°59 N, 98°49 E and altitude 2560 m. The PSNM has taken data since 2007 August, and since 2007 December 9 it has operated with a complete set of 18 BP-28 (¹⁰BF₃) neutron-sensitive proportional counter tubes with the standard NM64 configuration (18NM64). The 18 tubes are arranged in a single row to reduce edge effects. Further details are available from Ruffolo et al. (2016). At this altitude, the count rate is about 2.2 million counts hr⁻¹, providing a statistical accuracy better than 0.1% in hourly rates. The present study considers monthly or Carrington (solar) rotation (CarRot) averages (with truly negligible statistical uncertainty) from 2007 December 9 to 2017 January 7. At the end of that period, the apparent cutoff rigidity was 17.3 GV, and the effective vertical cutoff rigidity was 16.7 GV. Our calculated response function implies a median primary GCR rigidity of 35 GV and a 10th–90th percentile range of 18–155 GV (Ruffolo et al. 2016).

According to standard practice, PSNM data have been corrected for variations in atmospheric pressure, so the pressure-corrected count rate is $C_{\text{pcorr}}(t) = C(t)\exp[\beta(p - p_0)]$, where $C(t)$ is the measured count rate, $\beta = 0.641\% \text{ hPa}^{-1}$ is the

⁵ <http://cidas.isee.nagoya-u.ac.jp/WDCCR>

⁶ <http://nmdb.eu>

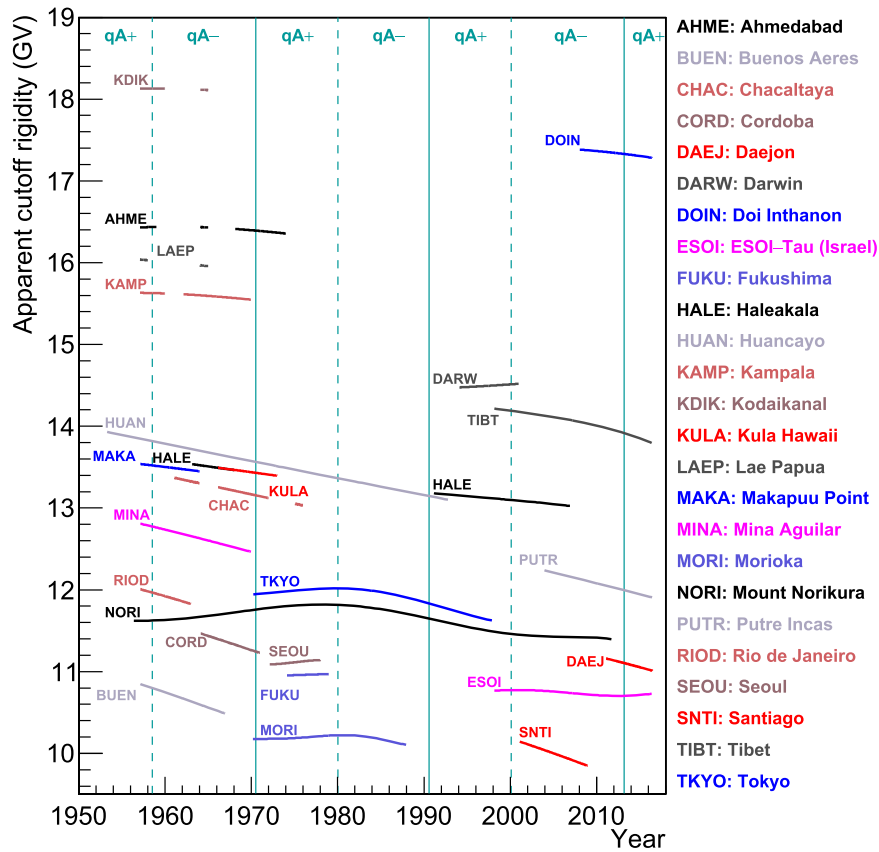


Figure 1. Time periods and apparent cutoff rigidities above 9.5 GV for which NM data are available. Reversals in solar magnetic polarity near sunspot maxima are indicated by vertical lines. Since NMs are sensitive to relativistic GCR ions, positive magnetic polarity $A > 0$ results in qA -positive solar modulation, and $A < 0$ results in qA -negative solar modulation. Among the few time series spanning a solar cycle, the NM data from Doi Inthanon have the highest apparent cutoff rigidity (over 17 GV) and are the focus of this study.

pressure coefficient determined from the data, and $p_0 = 750.6$ hPa is a reference pressure. Furthermore, Ruffolo et al. (2016) showed that the leader fraction L (inverse of mean multiplicity) derived from the neutron time delay distributions at Doi Inthanon is related not only to the atmospheric pressure but also to the surface water vapor pressure E_w . Because of the significantly different meteorological and environmental conditions during the dry and rainy seasons at this tropical location, for which the E_w values between ~ 1 and 15 hPa serve as a proxy, this effect is characterized by a yearly wave in L . Now considering the pressure-corrected count rate (see their Figure 5(a)), a hint of a yearly wave is again visible, though with a much smaller amplitude. Hence, in the present work, we have investigated the effect of surface water vapor on the PSNM count rate.

Following Ruffolo et al. (2016), we used data at 6 hr time intervals from the Global Atmospheric Data Assimilation⁷ to calculate E_w values for a pressure of 750 hPa (close to the mean pressure at the Doi Inthanon NM) at neighboring 1° grid points and interpolate to the location of Doi Inthanon. Figure 2 shows the pressure-corrected PSNM count rate $C_{P\text{corr}}$ for each 6 hr interval from 2007 December 9 to 2017 January 28 plotted as a function of E_w . There is a small but noticeable trend of decreasing count rate with increasing E_w , which is more apparent when considering the mean count rate in each 1 hPa interval of E_w (solid circles in Figure 2). While the fitting function used previously for the leader fraction by Ruffolo et al. (2016) was nonlinear in E_w , in this case, the mean count

rate is well modeled as a linear function of E_w . We therefore determine a pressure- and water vapor-corrected count rate $C_{PW\text{corr}}$ by dividing out the best-fit linear trend,

$$C_{PW\text{corr}}(t) = C_{P\text{corr}}(t) \frac{1 - aE_{w,0}}{1 - aE_w(t)}, \quad (1)$$

where the correction coefficient from the linear fit is $a = (2.05 \pm 0.19) \times 10^{-4} \text{ hPa}^{-1}$ and the reference value $E_{w,0} = 6.3$ hPa as in Ruffolo et al. (2016). We estimate the peak-to-peak annual wave in $C_{P\text{corr}}$ as the change when E_w varies from 1 to 15 hPa, which is 0.3%. The PSNM has observed a total peak-to-peak solar modulation of about 3%, so this correction is nonnegligible. In the remainder of this work, we only use $C_{PW\text{corr}}$, the pressure- and water vapor-corrected count rate.

3. Solar Modulation Observed by PSNM

3.1. Overview

Figures 3(a) and (b) present the count rates averaged over CarRots from the PSNM at Doi Inthanon and the McMurdo NM, respectively. The data reported in this work cover the period from 2007 December 9 (start of full operation at Doi Inthanon) to 2017 January 7 (closure of the McMurdo NM station). Both count rates have been corrected for pressure variation, and the Doi Inthanon count rate has also been corrected for water vapor variation as described in the previous section. Note that for clarity of presentation, the two count rates are shown on very different scales; the peak-to-peak solar

⁷ See <http://ready.arl.noaa.gov/gdas1.php>.

modulation at Doi Inthanon was $\approx 3\%$, whereas that at McMurdo was $\approx 15\%$. It is a well-known property of solar modulation that it is weaker at higher rigidity, and indeed, it may be negligible at $\gtrsim 50$ GeV.

In addition to the long-term process of solar modulation, the GCR flux undergoes Forbush decreases, which are temporary decreases due to the passage by Earth of a coronal mass ejection (CME) or other structures (Forbush 1937). A Forbush decrease typically involves a flux decrease over hours or days followed by a recovery over some days. When we average NM count rates over CarRots, the time profile of a Forbush decrease is not visible, yet the average count rate can be affected by the event. To check whether our understanding of the global heliospheric process of solar modulation is affected by the local effects of CME passage, which may have a different influence on the GCR spectrum, we show the NM count rates when time periods with a strong Forbush decrease are either included (light blue points) or excluded (black points). Here a strong Forbush decrease is defined as an event with a McMurdo count rate decrease of $>5\%$. Actually, for most months, there were no effects of strong Forbush decreases (i.e., the light blue and black points coincide), and none of the qualitative conclusions of our study depend on whether they are included or excluded. In subsequent sections, we exclude the times with strong Forbush decreases. Indeed, it is fascinating that even with strong Forbush decreases excluded, both NM count rates show numerous short-term modulation features that last only 1–2 CarRots. We speculate that these may be associated with CME-related structures that do not directly pass the Earth, yet as they propagate outward, they can affect solar modulation, e.g., by forming merged interaction regions or global merged interaction regions (Burlaga et al. 1983). These short-term features are stronger and more frequent during times of strong solar activity, supporting the idea that they are related to CMEs. Note also that these short-term variations affected the GCR spectrum near sunspot maximum at high rigidity (at Doi Inthanon) more strongly than solar modulation.

From late 2007, the two NM count rates increased until reaching a maximum in late 2009. This GCR maximum followed a very low minimum in the mean total sunspot number (shown in Figure 3(c)), as reported by WDC-SILSO, Royal Observatory of Belgium, Brussels,⁸ with some delay, as expected for solar effects to propagate throughout the heliosphere. Indeed, this time period witnessed the highest GCR flux in over 60 yr, since continuous NM measurements have been available (McDonald et al. 2010; Mewaldt et al. 2010; Oh et al. 2013). After the GCR maximum, both NM count rates initially decreased rather rapidly, until 2010 April. From 2007 December to 2010 April, the count rate variations at Doi Inthanon and McMurdo were well correlated.

Figure 3 also shows time series of the HCS tilt angle, solar wind speed, and IMF strength. These quantities could have causal effects on solar modulation, and detailed comparisons of their variations with NM count rate variations will be shown in later sections. For now, we discuss broad-brush comparisons for different time periods. At the time of the GCR maximum in 2009, all three quantities were near their minimum values. However, the IMF strength was nearly constant during the preceding period when the two NM count rates rose to their maxima, whereas the tilt and solar wind speed were

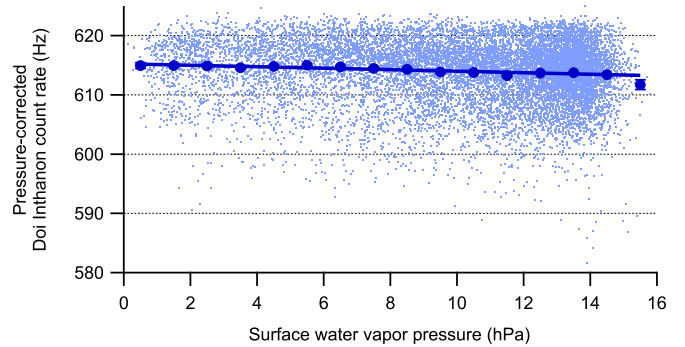


Figure 2. Scatter plot of the pressure-corrected NM count rate at Doi Inthanon as a function of surface water vapor pressure E_w for each 6 hr interval from 2007 December 9 to 2017 January 28, along with the mean count-rate values for 1 hPa intervals in E_w (solid circles). The linear fit to those mean values (solid line) was used for water vapor correction of the count rate.

anticorrelated with NM count rates during the GCR flux increase and initial decrease.

Then, starting in mid-2010, as the sunspot number and tilt angle generally increased, distinct patterns developed in the two NM count rates. Aside from the temporary decreases of 1–2 CarRots, as mentioned earlier, the McMurdo count rate (at low cutoff rigidity) generally decreased until early 2015, whereas the Doi Inthanon count rate (at high cutoff rigidity) remained relatively stable up to mid-2014. We call this the “plateau” period for Doi Inthanon. Then, in late 2014, the Doi Inthanon count rate underwent a sharp decrease to near its minimum value. Doi Inthanon observed a cosmic-ray maximum a few CarRots later than McMurdo and a cosmic-ray minimum a few CarRots earlier, so the total time from maximum to minimum was several months shorter.

Starting in 2015, both the Doi Inthanon and McMurdo count rates generally increased along with the general decline in the sunspot number, tilt angle, and IMF. In contrast, the solar wind speed was nearly constant (with significant fluctuations) and did not serve as a good predictor of solar modulation. During this period, the count rate variations at Doi Inthanon and McMurdo were again well correlated.

It is important to note that the Sun’s polar magnetic fields underwent polarity reversals during 2012–2014, during the Doi Inthanon plateau period, which was also a time of high sunspot number and high tilt angle, i.e., conditions near solar maximum. Figure 3(d) shows the solar polar field strengths as measured by the Wilcox Solar Observatory (WSO).⁹ The 20 nHz low-pass-filtered values are used here. Both north (B_n ; red) and south (B_s ; blue) polar magnetic field observations are presented, as well as their average $B_a = (B_n - B_s)/2$ (black). The zero crossing of the average value in 2013 March (black vertical line) is a useful indicator of the overall solar magnetic polarity reversal from $A < 0$ to $A > 0$. In any case, the complete sequence of polarity reversals lasted for about 2 yr, with three reversals near the North Pole from 2012 June to 2014 September and a single reversal near the South Pole in 2013 July.

Note that this 4 yr plateau started about 2 yr before the sequence of polarity reversals at the solar poles as observed by WSO (Figure 3(d)) and lasted until the end of the sequence. Thus, we see that this plateau period is not determined by the

⁸ See <http://sidc.be/silso/home>.

⁹ See <http://wso.stanford.edu/Polar.html>.

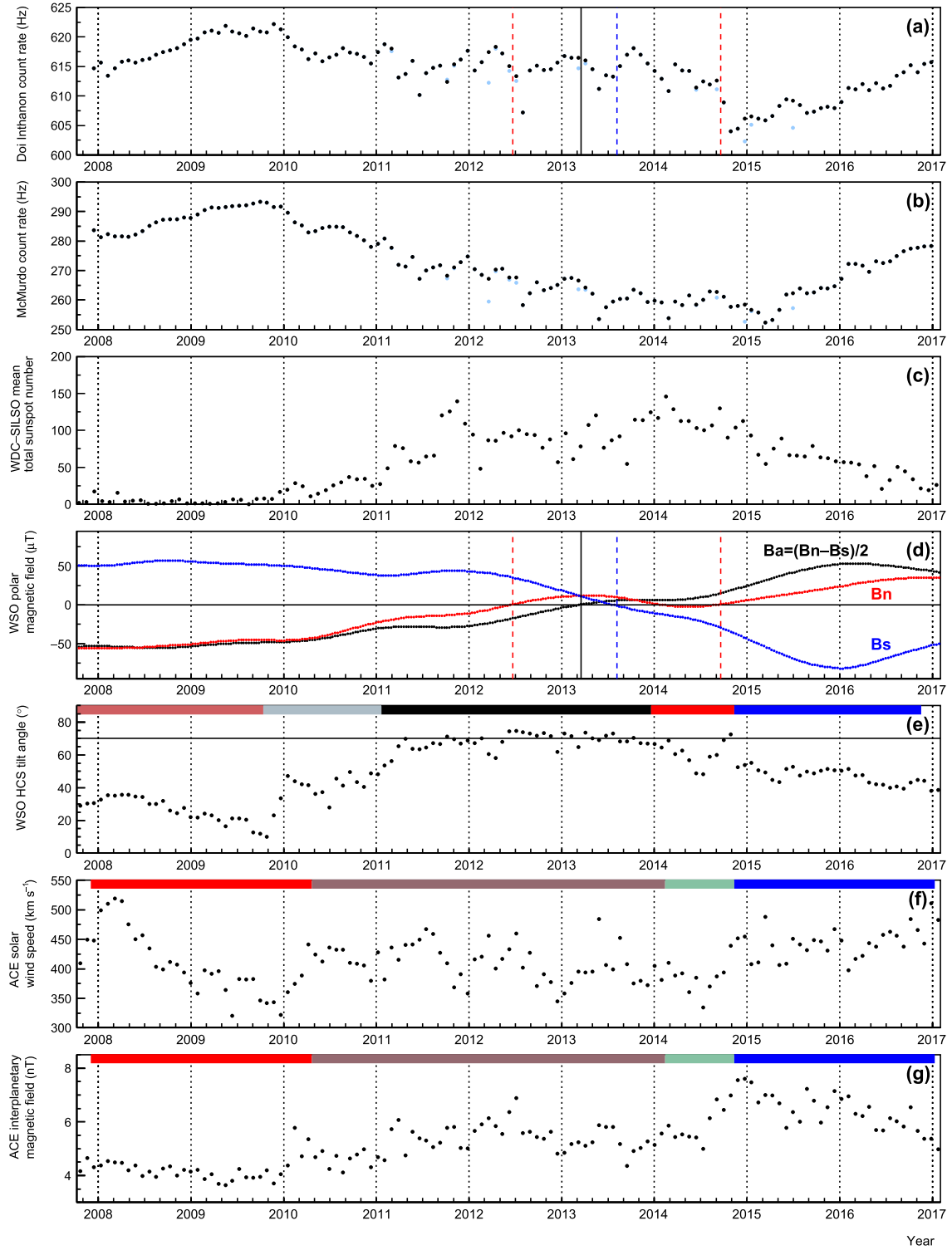


Figure 3. Time series from late 2007 to early 2017 of (a) the pressure- and water vapor-corrected NM count rate at Doi Inthanon, averaged over CarRots; (b) the pressure-corrected NM count rate at McMurdo, averaged over CarRots; (c) the WDC-SILSO monthly mean total sunspot number; (d) the WSO solar polar magnetic fields (traces explained in the text); (e) the WSO HCS tilt angle, averaged over CarRots; (f) the ACE/SWEPAM solar wind speed, averaged over CarRots; and (g) the ACE/MAG IMF strength, averaged over CarRots. In (a) and (b), fluxes before and after removing periods of strong Forbush decreases are shown by light blue and black points, respectively. Vertical lines in panels (a) and (d) indicate the first and last zero crossings of B_n (red) and the zero crossings of B_s (blue) and B_a (black). The horizontal line in panel (e) indicates a tilt angle of 70° ; values above this line should be considered as lower limits. The color code over panel (e) indicates time periods of distinct tilt angle levels or trends, as used in Figure 6, while another color code over panels (f) and (g) indicates time periods of distinct levels or trends in interplanetary parameters, as used in Figures 4, 7, and 8. A key point is that during the years 2011–2014, corresponding to very high tilt angles, the NMs at Doi Inthanon (at high cutoff) and McMurdo (at low cutoff) exhibited distinct patterns of solar modulation, whereas before and after that time (for lower tilt angles), the two NM count-rate variations were well correlated.

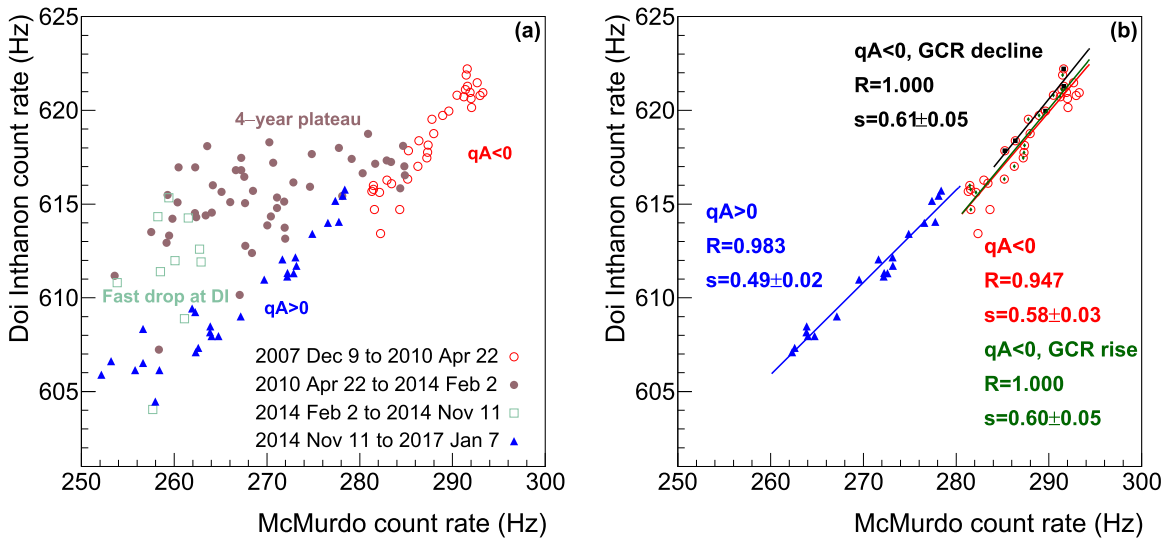


Figure 4. (a) Scatter plot of NM count rates at Doi Inthanon (high cutoff rigidity) and McMurdo (low cutoff rigidity) from 2007 December to 2017 January. (b) Same plot but restricted to the earliest and latest time periods, for low solar activity and an HCS tilt angle $\lesssim 50^\circ$; at such times, the two stations' count rates have a linear relationship, and the slope depends on solar magnetic polarity as found by Nuntiyakul et al. (2014). For a higher tilt angle, from mid-2010 to late 2014, there is no longer a linear relationship, as Doi Inthanon has a pattern of solar modulation distinct from that observed at low cutoff.

solar polarity reversal alone. The plateau period corresponds more closely to the period of very high tilt angle.

3.2. Comparison of PSNM Data with Other NM Data

There have been previous comparisons of the count rates of fixed NMs located at different rigidity cutoffs in order to study solar modulation over long periods. Its detailed evolution over previous solar cycles was analyzed by Popielawska & Simpson (1991) and Popielawska (1992, 1995) with NM data from Climax (~ 3 GV) and Huancayo (~ 13 GV) and from Kiel (~ 2.3 GV) and Tsumeb (~ 9.3 GV). Here we compare PSNM data from Doi Inthanon with NM data from McMurdo and examine how well a modulation parameter inferred from other NM data can be used to describe solar modulation as observed by the PSNM at high cutoff rigidity.

3.2.1. Comparison with McMurdo NM

Variations of the count rate at an apparent rigidity cutoff of ~ 17.3 GV, as observed at Doi Inthanon, show interesting behavior with respect to variations observed at McMurdo with an atmospheric cutoff at ~ 1 GV. The basic features were described in Section 3.1; here we compare the Doi Inthanon and McMurdo NM count rates in more detail. Figure 4(a) presents the scatter plot of the count rates (in Hz) at Doi Inthanon and McMurdo from the start of PSNM operations to the end of operations at McMurdo. Each marker is an average value per CarRot, and the marker color corresponds to the time period as indicated in Figures 3(f) and (g), i.e., from 2007 December to 2010 April, while the Doi Inthanon and McMurdo count rates were well correlated (red open circles); during the Doi Inthanon plateau (brown filled circles); during the fast drop at Doi Inthanon (green open squares); and during the GCR rise (blue filled triangles), when the two count rates were again well correlated.

Note that for clarity of presentation, the Doi Inthanon scale is again expanded relative to the McMurdo scale, with the full scale of solar modulation at Doi Inthanon corresponding to a much smaller percentage increase than that at McMurdo. As a

result, the temporary decreases during 1–2 CarRots, as noted earlier, have a strong relative impact on modulation at Doi Inthanon and cause a large vertical scatter in the graph during the 4 yr plateau period. As previously mentioned, the 4 yr plateau is followed by a transition period with a very fast drop in the count rate at Doi Inthanon. This decrease represents roughly half of the amplitude of the modulation observed during the full solar cycle at Doi Inthanon. In fact, a similar percentage decrease is present in the observations at McMurdo during the same time period, but that represents a smaller portion of the full modulation there.

During times of low tilt angle, the count rates at Doi Inthanon and McMurdo were linearly related. Figure 4(b) shows the linear trends between the count rates for such periods, i.e., from 2007 December 9 to 2010 April 22 for $qA < 0$ (red circles) and from 2015 July 14 to 2017 January 7 for $qA > 0$ (blue triangles). When $qA < 0$, the value of the slope was the same (within the uncertainties) during the GCR rise (green diamonds) and decline (black squares). After the period of high tilt angle near solar maximum, during which no clear correlation between the count rates was observed, the linear trend returned after the last large Forbush decrease, now with $qA < 0$. However, the slope of Doi Inthanon versus McMurdo count rate variations was lower when $qA > 0$ (slope ≈ 0.5) than when $qA < 0$ (slope ≈ 0.6). This confirms the linear trends between count rates at different cutoff rigidity, as observed from latitude surveys during the previous solar cycle and polarity reversal (Nuntiyakul et al. 2014), except for the year 2000 near solar maximum. The change in slope with solar magnetic polarity reversal, as reported here, is in the same direction as found in that work.¹⁰

¹⁰ Due to a misunderstanding of the units, the McMurdo NM count rates reported by Nuntiyakul et al. (2014) should all be multiplied by 5/3, and the regression coefficients of the shipborne monitor versus the McMurdo NM count rate should all be multiplied by 3/5. The qualitative conclusions of that work remain unchanged. Note also that it is difficult to compare absolute values of the slope from that work with those in the present work because the count rates depend on the number of counter tubes, altitude, tube configuration, dead time, and surrounding structure (Mangeard et al. 2016b).

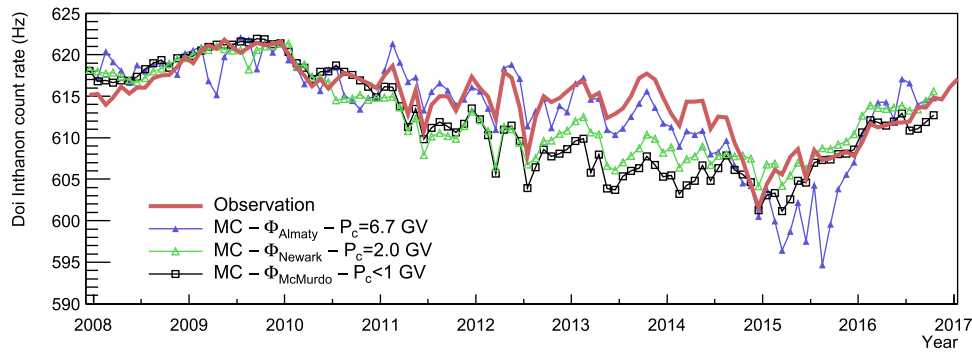


Figure 5. Comparison of the measured NM count rate at Doi Inthanon (thick red line) with simulated count rates for Doi Inthanon as estimated using yield functions from Monte Carlo simulation (Mangeard et al. 2016b) and a model GCR spectrum based on the LIS from Ghelfi et al. (2016, 2017a) and the force-field approximation. This approximation used solar modulation parameters estimated from NMs at different rigidity cutoffs, i.e., Almaty (blue line), Newark (green line), and McMurdo (black line), according to the method presented by Ghelfi et al. (2017b). The simulated count rates were normalized to that of the PSNM at the cosmic-ray maximum (2009 December). During times of high tilt angle, the solar modulation parameter time series based on NMs at low cutoff provide a poor description of solar modulation at Doi Inthanon at high cutoff, indicating that a model with a single empirical modulation parameter does not accurately describe solar modulation over a wide rigidity range.

3.2.2. Inadequacy of Modulation Parameter Derived from Low Cutoff NM Data

Recently, Ghelfi et al. (2017b) provided time series of empirical solar modulation parameters Φ_{NM} derived using data from a single NM.¹¹ They based their calculations on (i) the nonparametric proton and helium local interstellar spectra (LIS) determined from recent direct measurements with high statistics from balloon experiments (BESS, BESS-Polar) and spacecraft data (PAMELA, AMS-02; Ghelfi et al. 2016, 2017a), (ii) the force-field model of solar modulation (Gleeson & Axford 1968), and (iii) NM data from the station of interest. Time series of Φ_{NM} up to 2016 October were tabulated for various NMs located at different geomagnetic cutoffs.

Here we use such time series of Φ_{NM} to infer the GCR spectrum at Earth and yield functions for the PSNM based on Monte Carlo simulation results to generate simulated count rates for Doi Inthanon, which we then compare with the measured PSNM count rate (Figure 5). We employ monthly time series of Φ_{NM} from three NMs: McMurdo (Antarctica; $P_c \sim 1$ GV), Newark (USA; $P_c = 2.0$ GV), and Almaty (Kazakhstan; $P_c = 6.7$ GV). Note that some anomalous variations for Almaty in 2015 are not understood and not correlated with variations for other NM stations. For each of these three stations, we derive the time series of the spectra of protons and helium nuclei at the top of the atmosphere (based on the work of Ghelfi et al. 2016, 2017a, 2017b). We then apply the simulated yield functions from Mangeard et al. (2016b) to determine the count rates expected for the PSNM. We aim to examine the accuracy of a force-field model that uses a single empirical modulation parameter to describe solar modulation of the GCR spectrum over a wide rigidity range, given that models such as this are widely used in the community.

The comparison of the simulated count rates (thin lines) with the PSNM observations at Doi Inthanon (thick red line) is presented in Figure 5. The simulated count rates were normalized to that of the PSNM at solar minimum (2009 December). We see that that solar modulation of the PSNM count rate, i.e., the variation over the solar activity cycle, is poorly described by the force-field model that used a modulation parameter based on McMurdo data. This is

consistent with our observation of a different pattern of solar modulation at McMurdo and Doi Inthanon. Interestingly, the simulated PSNM count rates based on modulation parameters for Newark and Almaty, which are located at intermediate geomagnetic cutoffs, are usually bracketed by the simulated PSNM count rate based on McMurdo data (black line) and the actually observed PSNM count rate (except during the time period in 2015 with anomalous Almaty count rates). Thus, the accuracy of the force-field model systematically increases when the modulation parameter is based on NM data with cutoff rigidity closer to the value of interest. In particular, we can infer that solar modulation at Almaty, at $P_c = 6.7$ GV, is qualitatively (physically) more similar to that at Doi Inthanon than is solar modulation at low cutoff rigidity. Note also that the time lag in the modulation during the period of high solar activity increases with the cutoff. Finally, the main conclusion is that a force-field model with a single solar modulation parameter cannot reproduce the observations made at multiple rigidity cutoffs (~ 1 – 17 GV) over a full solar cycle.

3.3. Comparison with Heliospheric Measurements

In this section, we compare measurements of heliospheric parameters (HCS tilt angle, solar wind speed, and IMF) with NM count rates at Doi Inthanon and McMurdo, where all quantities are averaged over CarRots. Tables 1–3 show the correlation coefficient between each parameter and NM count rate as recorded after a delay time of 0–3 CarRots, in order to allow for the possibility that a heliospheric parameter value as measured near the Sun or Earth will propagate outward with the solar wind and then affect solar modulation as indicated by the NM count rates. Correlation coefficients are shown for all data and for four to five shorter time periods (color-coded in Figure 3) corresponding to phases of solar modulation in which there could be different relationships between the data. When considering the tilt angle, we found very different behaviors depending on the value of the tilt angle, so the time periods shown in Table 1 are used for the tilt angle data, which are compared with the NM data after the appropriate delay, to the extent that NM data were available from both Doi Inthanon and McMurdo. For the solar wind speed and magnetic field, the comparison depends on the values of the NM count rates, so the time periods shown in

¹¹ See <http://lpsc.in2p3.fr/crdb>.

Table 1
Correlation Coefficient between HCS Tilt Angle and NM Count Rates

Period	Polarity	Doi Inthanon NM				McMurdo NM			
		Tilt-to-NM Delay (CarRots)				Tilt-to-NM Delay (CarRots)			
		0	1	2	3	0	1	2	3
All		-0.42	-0.45	-0.47	-0.46	-0.78	-0.81	-0.83	-0.84
Start to 2009 Oct 13	$A < 0$	-0.91	-0.87	-0.84	-0.79	-0.93	-0.92	-0.86	-0.79
2009 Oct 13 to 2011 Jan 1	$A < 0$	-0.55	-0.62	-0.70	-0.66	-0.68	-0.75	-0.81	-0.73
2011 Jan 1 to 2013 Dec 19	$A < 0 \rightarrow A > 0$	-0.42	-0.26	-0.05	-0.19	-0.55	-0.51	-0.50	-0.34
2013 Dec 19 to 2014 Nov 11	$A > 0$	-0.45	-0.39	-0.29	-0.00	-0.33	-0.50	-0.73	-0.58
2014 Nov 11 to End	$A > 0$	-0.73	-0.75	-0.77	-0.78	-0.65	-0.67	-0.71	-0.74

Table 2
Correlation Coefficient between Solar Wind Speed (SWS) and NM Count Rates

Period	Polarity	Doi Inthanon NM				McMurdo NM			
		SWS-to-NM Delay (CarRots)				SWS-to-NM Delay (CarRots)			
		0	1	2	3	0	1	2	3
All		-0.41	-0.36	-0.28	-0.19	-0.10	-0.08	-0.05	0.00
2007 Dec 9 to 2010 Apr 22	$A < 0$	-0.88	-0.82	-0.71	-0.67	-0.88	-0.81	-0.73	-0.65
2010 Apr 22 to 2014 Feb 2	$A < 0 \rightarrow A > 0$	-0.25	-0.27	-0.20	-0.05	0.14	0.15	0.15	0.17
2014 Feb 2 to 2014 Nov 11	$A > 0$	-0.64	0.11	0.24	0.56	-0.56	-0.41	-0.67	-0.13
2014 Nov 11 to 2017 Jan 7	$A > 0$	0.31	0.25	0.46	0.51	0.43	0.26	0.40	0.42

Table 3
Correlation Coefficient between IMF and NM Count Rates

Period	Polarity	Doi Inthanon NM				McMurdo NM			
		IMF-to-NM Delay (CarRots)				IMF-to-NM Delay (CarRots)			
		0	1	2	3	0	1	2	3
All		-0.85	-0.84	-0.78	-0.75	-0.75	-0.74	-0.71	-0.68
2007 Dec 9 to 2010 Apr 22	$A < 0$	-0.48	-0.54	-0.53	-0.65	-0.51	-0.60	-0.63	-0.66
2010 Apr 22 to 2014 Feb 2	$A < 0 \rightarrow A > 0$	-0.55	-0.57	-0.26	-0.15	-0.56	-0.55	-0.46	-0.33
2014 Feb 2 to 2014 Nov 11	$A > 0$	-0.78	-0.54	-0.45	-0.37	0.02	-0.10	-0.15	-0.14
2014 Nov 11 to 2017 Jan 7	$A > 0$	-0.69	-0.58	-0.38	-0.27	-0.61	-0.53	-0.45	-0.37

Tables 2–3 are used for the NM count rate data, which are compared with the solar wind speed and IMF data at an earlier time to give the appropriate delay. In all tables, particularly strong correlations (of absolute value 0.70 or greater) are indicated in bold type. For each parameter, an optimal delay time is identified, which is used when comparing with NM count rates in Figures 6–8.

3.3.1. HCS Tilt Angle

The “classic” average tilt angle (Hoeksema 1995) computed from WSO observations¹² is shown in Figure 3(e), which gives information about the HCS near the Sun only. Its correlation with the inclination of the HCS further in the heliosphere is expected to be better during or near solar minimum. Near solar maximum, the fast variation of conditions near the Sun makes it difficult to directly extrapolate from the tilt angle to the HCS properties at a long distance from the Sun. Furthermore, the tilt angle data are accompanied by a notice that values above 70° (indicated by a horizontal line in Figure 3(e) and vertical lines

in Figure 6) should be considered as lower limits, so near solar maximum, the tilt angle is not well measured.

Table 1 presents the correlation coefficients between the count rates at Doi Inthanon and McMurdo and the tilt angle for five time periods based on the value of the tilt angle and the solar activity and for four delays, 0–3 CarRots. A delay time of 2 CarRots provides the greatest number of strong (>70%) correlation coefficients with the NM data during various time periods. Figure 6 presents the correlations between the count rates (average per CarRot) at Doi Inthanon and McMurdo with the tilt angle after the optimal delay of 2 CarRots was applied to the count rates. Note that the time periods indicated in the legend refer to times of NM data to simplify the comparison with other figures.

During the initial period of low solar activity with a tilt angle below 50° at negative solar magnetic polarity ($A < 0$), Figure 6 shows that both NM count rates were strongly anticorrelated with the tilt angle, either during the GCR rise (open red circles) or GCR decline (solid gray circles). For McMurdo, the trends during the $A < 0$ GCR rise and decline at such low tilt angles are similar, and it could be said that the tilt angle is an excellent predictor of the McMurdo count rate,

¹² See <http://wso.stanford.edu/Tilts.html>.

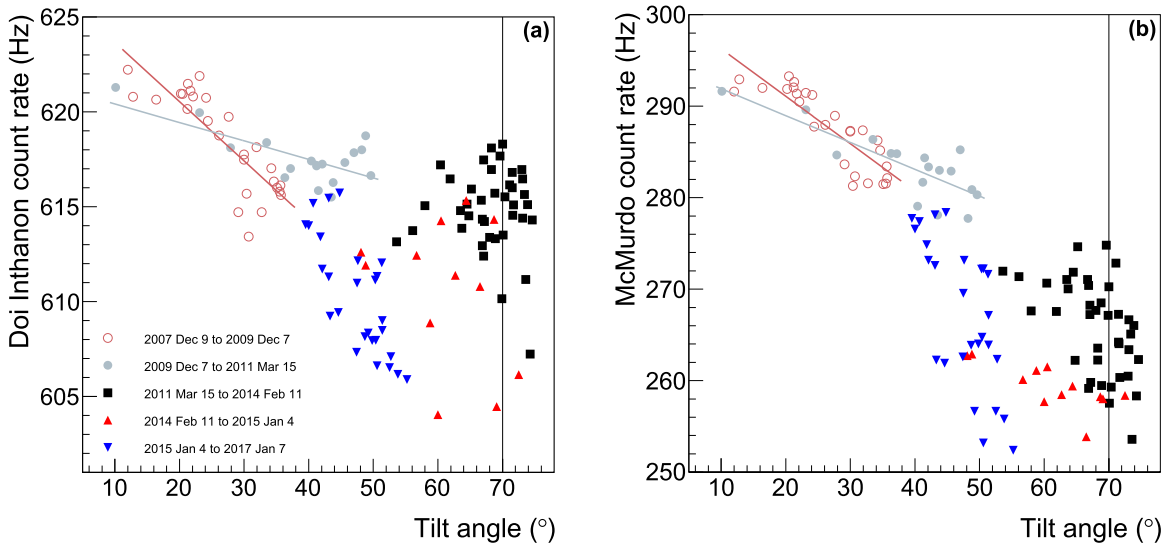


Figure 6. Scatter plots of the CarRot-averaged count rates at (a) Doi Inthanon and (b) McMurdo with the WSO HCS tilt angle. NM count rates were compared with the tilt angle from 2 CarRots earlier to maximize the correlation factors; the dates indicated in (a) refer to the dates of NM data.

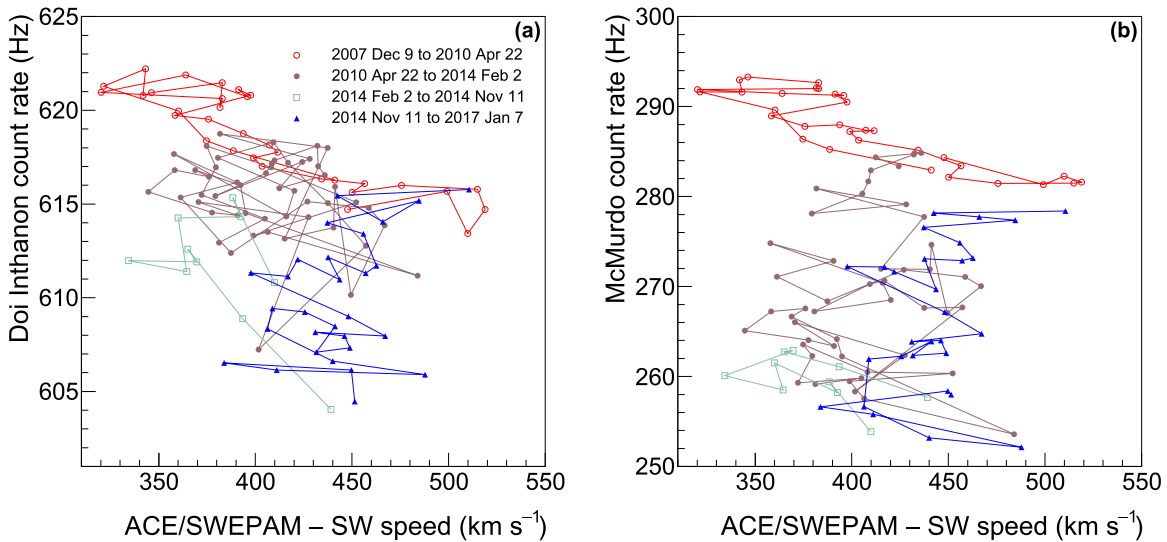


Figure 7. Scatter plots of the CarRot-averaged count rates at (a) Doi Inthanon and (b) McMurdo with the contemporaneous *ACE*/*SWEPAM* solar wind speed, averaged over CarRots.

as expected from drift models (Jokipii & Thomas 1981; Strauss et al. 2012). Then, at a tilt angle $>50^\circ$ and during the period of solar polarity reversals (black squares), the McMurdo count rate was suddenly lower, and at $>60^\circ$, the count rate is apparently uncorrelated with the measured tilt angle. After the period of polarity reversals, during 2014 February to 2015 January (red triangles), the McMurdo count rate recovered only slightly with decreasing tilt angle, and it recovered much more rapidly in the final time period (blue triangles).

While the pattern for Doi Inthanon was similar at low solar activity, the pattern changed in mid-2010 at the start of the “plateau” count rate, as the tilt angle exceeded 40° . During the third and fourth time periods, there was little or no correlation between the tilt angle and the Doi Inthanon count rate. During the final period, when the tilt angle was below $\sim 50^\circ$, the Doi Inthanon count rate increased rapidly while the tilt angle was decreasing, as seen for McMurdo.

3.3.2. Solar Wind Speed

Next, we consider the solar wind speed, as measured by the Solar Wind Electron, Proton, and Alpha Monitor¹³ (SWEPAM) on the *Advanced Composition Explorer* (ACE) spacecraft (located at the L1 Lagrange point) as the second heliospheric measurement that we compared to the count rates at Doi Inthanon and McMurdo. The optimal delay is zero, for which the overall correlation factor between the count rates and the solar wind is -0.41 and -0.10 for Doi Inthanon and McMurdo, respectively (see Table 2), indicating a poor correlation over the course of the solar modulation cycle.

Figures 7(a) and (b) present scatter plots between the solar wind speed and the count rates (averaged per CarRot) at Doi Inthanon and McMurdo, respectively, which confirm the poor overall correlation. However, during the initial period of low solar activity and negative solar polarity, there was a strong

¹³ See http://www.srl.caltech.edu/ACE/ASC/level2/swepam_l2desc.html.

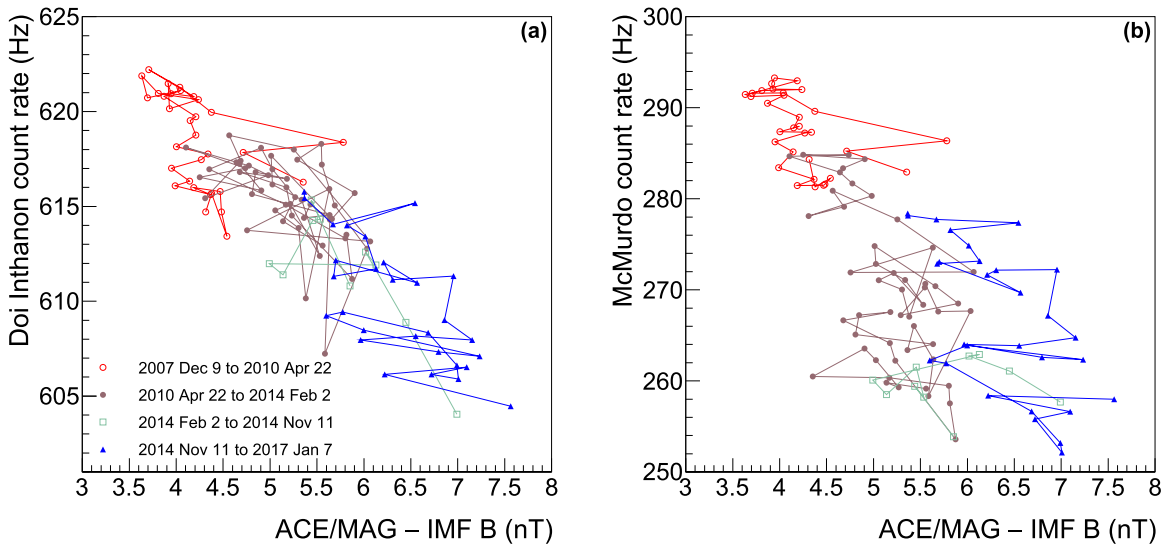


Figure 8. Scatter plots of the CarRot-averaged count rates at (a) Doi Inthanon and (b) McMurdo with the contemporaneous *ACE*/*MAG* IMF, averaged over CarRots.

anticorrelation for both stations, with a correlation coefficient of -0.88 for both NM stations. During the period of high solar activity, as well as the last period of low solar activity at positive solar magnetic polarity, the correlation with the solar wind speed at Earth was quite weak. This suggests that the underlying physics of the solar modulation during these time periods was not related to the local solar wind speed at Earth.

3.3.3. IMF

Finally, we consider the IMF, here defined as the mean magnitude of the magnetic field as measured by the Magnetic Field Experiment¹⁴ (*MAG*) on the *ACE* spacecraft, in comparison with NM count rates. The optimal delay time is zero, for which the overall correlation coefficient between the count rates and the solar wind is -0.85 and -0.74 for Doi Inthanon and McMurdo, respectively (see Table 3). Figure 3(f) presents the time series of the IMF during the period of interest of this work. Figures 8(a) and (b) present scatter plots between the IMF and the count rates (averaged per CarRot) at Doi Inthanon and McMurdo, respectively.

The McMurdo NM count rate, at low cutoff rigidity, has a strong overall anticorrelation with the IMF, but each individual time period has a substantially weaker correlation, and during the third time period (2014 February–November) there was virtually no correlation at all. Thus, the IMF serves to roughly indicate times of weak or strong solar modulation (i.e., low or high solar activity) but does not correlate well with the details of solar modulation as observed by a polar NM at low cutoff rigidity.

In contrast, the Doi Inthanon NM count rate, at high cutoff rigidity, has a very strong overall anticorrelation with the IMF that is also quite strong during the third period (2014 February–November). Among the heliospheric parameters we have considered, the IMF clearly has the best predictive power for the Doi Inthanon count rate, approximately following a common linear trend for all time periods of solar modulation. An exception is that during the first period, corresponding to $A < 0$ solar magnetic polarity and low tilt angle, when the tilt angle has a tighter correlation with the Doi Inthanon count rate.

During the second period, there was a nearly constant IMF and a corresponding plateau in the Doi Inthanon count rate, both of which were punctuated by short-term (1–2 CarRot) modulation events. Thus, the somewhat lower correlation coefficient during this time period does not indicate that the IMF fails to explain the Doi Inthanon count rate, because the nearly constant IMF is consistent with the 4 yr plateau in the count rate. For this solar modulation cycle, the IMF is clearly a key determinant of the high cutoff count rate.

4. Discussion

We have examined the first set of continuous NM data of the time-varying cosmic-ray flux at cutoff rigidity >16 GV over a cycle of solar modulation. We compared these data from Doi Inthanon from 2007 December 9 until 2017 January 7 with data over the same time period from McMurdo with an atmospheric cutoff rigidity of ~ 1 GV. During times when the HCS tilt angle was $\lesssim 50^\circ$, corresponding to periods of lower solar activity, the Doi Inthanon and McMurdo count rates were linearly related (see Figure 4). However, at other times, they exhibited distinct patterns of solar modulation, which we attribute to a difference in the dominant physical mechanism.

For latitude survey data from 1994 to 2007, either before or after survey year 2000 with maximum solar activity, Nuntiyakul et al. (2014) concluded that at a given cutoff rigidity, the mobile monitor count rate as a function of time was linearly correlated with the count rate at McMurdo. The slope of the linear relation was higher during the time period of $A < 0$ solar magnetic polarity. Our present results comparing the fixed NM data from Doi Inthanon and McMurdo are qualitatively consistent with the latitude survey results. (Note that PSNM data can be used to measure the slope of the linear relation much more precisely for continuous measurements with 18 NM counters at high altitude compared with three NM counters on a ship that only occasionally reached that cutoff rigidity.) One difference is that the time period of the present work includes a 4 yr period of high solar activity, from 2010 April to 2014 November, when the Doi Inthanon and McMurdo NM count rates were not well correlated, which roughly corresponds to a period of HCS tilt angle $\gtrsim 50^\circ$. In contrast, the time period of those latitude surveys included such

¹⁴ See http://www.srl.caltech.edu/ACE/ASC/level2/mag_l2desc.html.

a high tilt angle only during about 1999–2000 (McDonald et al. 2010), which is consistent with their need to exclude only survey year 2000 from their periods with linear relationships between NM count rates. In summary, the present study, as well as those of Nuntiyakul et al. (2014) and Popielawska (1992), paints a consistent picture that NM count rates at different cutoff rigidities are linearly related under conditions of low solar activity, with a slope that depends on the solar magnetic polarity.

Based on our finding of different patterns of solar modulation at low and high cutoff rigidity for over 4 yr at high solar activity and high HCS tilt angle, from 2010 April to 2014 November, it is perhaps unsurprising that a simple force-field model of solar modulation, with a single modulation parameter inferred from low cutoff NM data as commonly used to model solar effects on the GCR spectrum, fails to accurately describe solar modulation above a high cutoff rigidity ($\gtrsim 17$ GV). From Figure 5, we see that when using an empirical modulation parameter as a function of time based on data from low cutoff rigidity (~ 1 GV), the predicted solar modulation in the NM count rate (relative to the count rate maximum) from a force-field model was typically a factor of two greater than the modulation observed at Doi Inthanon during that 4 yr period. This indicates that a force-field model based on a single modulation parameter can fail to even approximately describe changes in the GCR spectrum during times of high solar activity and high tilt angle. On the other hand, during periods of low tilt angle, this force-field model performs reasonably well in simultaneously matching NM variations at low and high cutoff rigidity (keeping in mind that the traces in Figure 5 are all normalized at the solar minimum conditions of 2009 December), given an empirical modulation parameter based on data from one NM station. Apparently, the assumptions of the force-field model, such as spherical symmetry, a quasi-steady state, and a diffusion coefficient that increases linearly with rigidity, work reasonably well during relatively quiet solar conditions and are able to reproduce observed count-rate variations relative to the relevant solar minimum count rate. Note, however, that the effects of solar magnetic polarity on the GCR ion spectrum (Moraal et al. 1989; Nuntiyakul et al. 2014) are not included in the force-field model.

We have also examined the effects of three heliospheric parameters, the HCS tilt angle, solar wind speed, and IMF, on the Doi Inthanon and McMurdo NM rates. The tilt angle refers to the global tilt of the HCS near the Sun and thus might be expected to be relevant to the heliosphere as a whole. Furthermore, *Ulysses* observations have indicated that the radial magnetic flux $|B_r|r^2$ is independent of location in the inner heliosphere at a given time (Smith & Balogh 1995), so the IMF at any one point (e.g., near Earth) is indicative of the global IMF throughout much of the heliosphere. In contrast, the solar wind speed near Earth, even when averaged over CarRots, is a more local indicator of conditions near the solar equatorial plane. We find a weaker correlation between the local solar wind speed and solar modulation of NM count rates, which is consistent with the common view that solar modulation mainly reflects global conditions in the heliosphere.

Tables 1–3 show the correlation coefficient of each parameter as compared with each NM count rate as measured after a delay time of 0–3 CarRots, and the optimal comparisons are plotted in Figures 6–8. The Doi Inthanon count rate has a strong overall anticorrelation with the magnetic field (with no

delay), with an overall linear trend. The IMF is also the parameter that best correlates with the Doi Inthanon count for individual time periods, except during the initial period of low solar activity and $A < 0$ solar magnetic polarity, when the tilt angle (with 2 CarRot delay) provides a better correlation. During the GCR rise with $A > 0$ polarity in the final period (2014 November–2017 January), the correlation coefficients for the IMF and tilt angle are similar, but the slope of the GCR rise with decreasing tilt angle is so strong that the tilt angle has little predictive power during that time period. For the McMurdo count rate, the HCS tilt angle has a strong anticorrelation with the McMurdo count rate (especially after a delay of 2 CarRots) and is clearly better correlated than the other heliospheric parameters. However, the relation is not linear and exhibits hysteresis, as the McMurdo count rate follows a different track as a function of tilt angle after the solar magnetic polarity reversal.

The high NM count rates near solar minimum coincide with a very low value of the tilt angle, $\sim 10^\circ$ – 20° , in agreement with the calculation of Webber et al. (1990). Their model, which includes the effects of guiding center drifts and a wavy neutral sheet, describes a dependence of the solar modulation on the tilt angle during negative ($A < 0$) solar polarity. For this polarity, drift models generally predict that positively charged particles (including GCR ions) enter the inner solar system by drifting along the wavy path following the HCS current sheet (Jokipii & Thomas 1981; Strauss et al. 2012). Thus, a small tilt angle implies a shorter path length and less scattering on the way to Earth, with a lower modulation effect and higher count rate in comparison to that at large tilt angle. Our observations confirm that at low tilt angle and negative solar polarity, the drift of the cosmic-ray particles in the IMF seems to affect NM count rates up to a cutoff rigidity of about 17 GV.

At high tilt angle, the Doi Inthanon count rate is poorly correlated with the tilt angle. On the other hand, there is a strong overall anticorrelation between the magnetic field strength and the Doi Inthanon count rate with a single linear trend for the entire solar modulation cycle. An inverse correlation between the IMF and the count rate of an NM at low cutoff (Mount Wellington) was previously observed and described in Cane et al. (1999). They also reported a loss of correlation in 1969, 1980, and 1990 near the periods of solar polarity reversal. For Doi Inthanon, at high cutoff rigidity, we observe that the IMF still has useful predictive power even near the periods of polarity reversal. In particular, the 4 yr plateau in the Doi Inthanon count rate from mid-2010 to mid-2014 (which is not observed in the McMurdo count rate) corresponded to a period in which the IMF fluctuated over a moderate value of about 4–6 nT with little or no long-term time variation. Then, shortly after the IMF began to increase rapidly in the latter half of 2014, rising from about 5 to 8 nT, the Doi Inthanon count rate underwent a sharp drop over about 2 months (2014 September–October) that accounted for about half of the total solar modulation over this cycle at Doi Inthanon. These results strongly suggest a causal link between changes in the IMF and the NM count rate at high cutoff rigidity, which is not observed at low cutoff.

We propose the following physical explanation of this distinct pattern of solar modulation of GCRs at high cutoff rigidity. Changes in the magnetic field strength over the solar cycle should result in changes in the GCR diffusion coefficient throughout the heliosphere, and GCRs of high rigidity should

have a much higher diffusion coefficient (Chhiber et al. 2017). As noted above, the usual expectation of the effect of guiding center drifts on GCR ions during $A < 0$ solar magnetic polarity is that they should follow a convoluted path as they drift inward along the HCS, with a path length that depends on the tilt angle. Our interpretation of the distinct pattern of solar modulation at Doi Inthanon is that GCRs of high rigidity can have a very large diffusion coefficient and can “short-circuit” the drift process along the wavy current sheet, providing a shorter path length and access to the inner heliosphere that depends mostly on the diffusion coefficient, and hence the IMF. This view is supported by Strauss et al. (2012), in particular the simulation results in their Figure 12, in which a higher diffusion coefficient is seen to short-circuit the very long path along a highly tilted current sheet, implying that modulation is insensitive to the exact value of a large tilt angle, as found in our observations. Then, the result shown in Figure 5, that a force-field model with a modulation parameter derived from low cutoff NM data is unable to explain high cutoff solar modulation near solar maximum, can be attributed to the very different trajectories and physical processes experienced by cosmic rays above a high cutoff rigidity. This different type of heliospheric propagation of GCRs with high rigidity and large diffusion coefficients can give rise to the distinct pattern of solar modulation that we observe at tilt angles $\gtrsim 50^\circ$ above a high cutoff rigidity (~ 17 GV).

This work was partially supported by the Thailand Research Fund award RTA5980003 and the US National Science Foundation awards PLR-1245939, PLR-1341562, and their predecessors. We acknowledge the NMDB database, founded under the European Union’s FP7 program (contract No. 213007), and the PIs of individual NMs for providing data. We thank David Maurin for useful discussions.

ORCID iDs

Pierre-Simon Mangeard  <https://orcid.org/0000-0003-4865-6968>

Paul Evenson  <https://orcid.org/0000-0001-7929-810X>

David Ruffolo  <https://orcid.org/0000-0003-3414-9666>

Alejandro Sáiz  <https://orcid.org/0000-0001-7771-4341>

References

- Abe, K., Fuke, H., Haino, S., et al. 2016, *ApJ*, 822, 16
 Adriani, O., Barbarino, G. C., Bazilevskaya, G. A., et al. 2011, *Sci*, 332, 69

- Adriani, O., Barbarino, G. C., Bazilevskaya, G. A., et al. 2013, *ApJ*, 765, 91
 Adriani, O., Barbarino, G. C., Bazilevskaya, G. A., et al. 2016, *PhRvL*, 116, 241105
 Aguilar, M., Aisa, D., Alpat, B., et al. 2015a, *PhRvL*, 114, 171103
 Aguilar, M., Aisa, D., Alpat, B., et al. 2015b, *PhRvL*, 115, 211101
 Asaoka, Y., Shikaze, Y., Abe, K., et al. 2002, *PhRvL*, 88, 051101
 Bazilevskaya, G. A., Usoskin, I. G., Flückiger, E. O., et al. 2008, *SSRv*, 137, 149
 Bieber, J. W., & Burger, R. A. 1990, *ApJ*, 348, 597
 Bieber, J. W., Clem, J., & Evenson, P. 1997, Proc. ICRC (Durban), 2, 389
 Burlaga, L. F., Schwenn, R., & Rosenbauer, H. 1983, *GeoRL*, 10, 413
 Caballero-Lopez, R. A., & Moraal, H. 2004, *JGRA*, 109, A01101
 Caballero-Lopez, R. A., & Moraal, H. 2012, *JGRA*, 117, A12103
 Cane, H. V., Wibberenz, G., Richardson, I. G., & von Rosenvinge, T. T. 1999, *GeoRL*, 26, 565
 Chhiber, R., Subedi, P., Usmanov, A. V., et al. 2017, *ApJS*, 230, 21
 Clem, J., & Evenson, P. 2009, *JGRA*, 114, A10108
 Clem, J. M., Bieber, J. W., Evenson, P., et al. 1997, *JGR*, 102, 26919
 Clem, J. M., Evenson, P., Huber, D., et al. 2000, *JGR*, 105, 23099
 Corti, C., Bindi, V., Consolandi, C., & Whitman, K. 2016, *ApJ*, 829, 8
 Forbush, S. E. 1937, *PhRv*, 51, 1108
 Forbush, S. E. 1954, *JGR*, 59, 525
 Garcia-Munoz, M., Meyer, P., Pyle, K. R., Simpson, J. A., & Evenson, P. 1986, *JGR*, 91, 2858
 Ghelfi, A., Barao, F., Derome, L., & Maurin, D. 2016, *A&A*, 591, A94
 Ghelfi, A., Barao, F., Derome, L., & Maurin, D. 2017a, *A&A*, 605, C2
 Ghelfi, A., Maurin, D., Cheminet, A., et al. 2017b, *AdSpR*, 60, 833
 Gieseler, J., Heber, B., & Herbst, K. 2017, *JGRA*, 122, 10964
 Gleeson, L. J., & Axford, W. I. 1968, *ApJ*, 154, 1011
 Hoeksema, J. T. 1995, *SSRv*, 72, 137
 Jokipii, J. R., Levy, E. H., & Hubbard, W. B. 1977, *ApJ*, 213, 861
 Jokipii, J. R., & Thomas, B. 1981, *ApJ*, 243, 1115
 Levy, E. H. 1976, *JGR*, 81, 2082
 Lin, Z., Bieber, J. W., & Evenson, P. 1995, *JGR*, 100, 23543
 Mangeard, P.-S., Ruffolo, D., Sáiz, A., et al. 2016a, *JGRA*, 121, 11620
 Mangeard, P.-S., Ruffolo, D., Sáiz, A., Madlee, S., & Nutaro, T. 2016b, *JGRA*, 121, 7435
 McDonald, F. B., Webber, W. R., & Reames, D. V. 2010, *GeoRL*, 37, L18101
 Mewaldt, R. A., Davis, A. J., Lave, K. A., et al. 2010, *ApJL*, 723, L1
 Mishev, A. L., Usoskin, I. G., & Kovaltsov, G. A. 2013, *JGRA*, 118, 2783
 Mitthumsiri, W., Seripienlert, A., Tortempun, U., et al. 2017, *JGRA*, 122, 7946
 Moraal, H., Potgieter, M. S., Stoker, P. H., & van der Walt, A. J. 1989, *JGR*, 94, 1459
 Nuntiyakul, W., Evenson, P., Ruffolo, D., et al. 2014, *ApJ*, 795, 11
 Oh, S., Bieber, J. W., Evenson, P., et al. 2013, *JGRA*, 118, 5431
 Popielawska, B. 1992, *P&SS*, 40, 811
 Popielawska, B. 1995, *JGR*, 100, 5883
 Popielawska, B., & Simpson, J. A. 1991, Proc. ICRC (Dublin), 3, 660
 Ruffolo, D., Sáiz, A., Mangeard, P.-S., et al. 2016, *ApJ*, 817, 38
 Simpson, J. A. 2000, *SSRv*, 93, 11
 Smith, E. J., & Balogh, A. 1995, *GeoRL*, 22, 3317
 Strauss, R. D., Potgieter, M. S., Büsching, I., & Kopp, A. 2012, *ApSS*, 339, 223
 Usoskin, I. G., Gil, A., Kovaltsov, G. A., Mishev, A. L., & Mikhailov, V. V. 2017, *JGRA*, 122, 3875
 Webber, W. R., Potgieter, M. S., & Burger, R. A. 1990, *ApJ*, 349, 634